GameStop Collectibles Dynamic Pricing Engine

Technical Solution Architecture & Implementation Plan

Prepared by: Ish Prasad | Staff AI/ML Engineer **Date:** April 2026 **For:** Ravi & Team — GameStop AI/ML Engineering

Executive Summary

GameStop's collectibles business (graded trading cards — Pokemon, sports, TCG) currently prices cards using daily batch scrapes from eBay, with no price history storage, no sentiment analysis, and no ML-based pricing. This document presents a complete technical architecture for a **dynamic pricing engine** that combines multi-source market data, social sentiment, ML price prediction with uncertainty quantification, and anomaly detection — all designed to be built in-house with open-source tools.

The core insight: Collectibles pricing is a *sparse, irregular, fat-tailed* prediction problem — not a standard time series. Most cards sell infrequently (1-10x per month), prices can 10x overnight on social virality, and the same card at different grades can differ 50x in value. The system must handle this heterogeneity with volume-aware routing, not a one-size-fits-all model.

GameStop's unique advantage: Physical store foot traffic + trade-in pricing data that no online competitor (PriceCharting, Alt, Card Ladder) has access to. This is the moat.

1. Data Architecture

1.1 Data Sources — Priority Ranked

PRIORITY	SOURCE	DATA TYPE	ACCESS METHOD	COST	UPDATE FREQUENCY
P0	eBay Marketplace Insights API	Sold listings (90 days)	Limited Release API — negotiate access	Negotiate	Event-driven (every 15 min)
P0	SoldComps API	eBay sold listings (365 days)	REST API	$59/mo (5,000 req/mo)	Every 15 min (cron)
P0	PriceCharting API	Aggregated prices (all time)	REST API (40-char token)	~$15-30/mo (Legendary tier)	Daily (2 AM cron)
P1	PSA Population Reports	Grade population counts	Web scrape (psacard.com/pop) or GemRate	Free (GemRate)	Daily (3 AM cron)
P1	CardHedger API	40M+ transactions, multi-platform	REST API	$49+/mo	Near real-time
P1	Reddit (old.reddit.com)	Social sentiment	Browser Rendering scrape (old.reddit.com)	Free (Cloudflare Workers)	Every 5 min (cron)
P2	Twitter/X API	Event detection, viral moments	Pay-per-use	~$500-1K/mo	**DEFERRED**
P2	TCGPlayer API	Pokemon/MTG market prices	Apply for access	Free (with approval)	Daily
P2	PokemonPriceTracker API	Graded Pokemon prices + pop	REST API	$99/mo (Business)	Daily
P3	GameStop internal	Trade-in data, foot traffic, inventory	Internal DB	Free	**DEFERRED** — requires integration

1.2 eBay Strategy — Legal & Practical

Do NOT scrape eBay directly. eBay explicitly bans AI agents and scraping in their Feb 2026 User Agreement. As a publicly traded company with an eBay business relationship, GameStop cannot afford the legal/business risk.

Instead: 1. **Negotiate Marketplace Insights API access** — GameStop's retail footprint gives leverage. This API returns sold items for last 90 days and is the gold standard. 2. **Use SoldComps as fallback** ($59/mo) — licensed eBay sold data via REST API, up to 240 results per request, 365 days history. 3. **Use CardHedger for multi-platform** ($49/mo) — aggregates eBay, Fanatics, Heritage Auctions, and more.

1.3 Price History Storage

GameStop doesn't store price history. This is the foundation everything else depends on.

Built on Cloudflare D1 (SQLite at the edge, zero cold starts, global replication, ~$5-20/mo at moderate scale):

```
Tables (all implemented):
├── card_catalog        — master card registry (id, name, set_name, set_year, card_number,
│                         category, player_character, team, rarity, image_url, pricecharting_id)
│                         Categories: pokemon, sports_baseball, sports_basketball,
│                         sports_football, sports_hockey, tcg_mtg, tcg_yugioh, other
├── price_observations  — every API price point
│   (card_id, source, price_usd, sale_date, grade, grading_company, grade_numeric,
│    sale_type, seller_id, bid_count, listing_url, is_anomaly, anomaly_reason)
│   Sources: ebay, soldcomps, pricecharting, cardhedger, tcgplayer, gamestop_internal
│   Dedup: unique index on (card_id, source, listing_url)
├── price_aggregates    — daily/weekly/monthly rollups per card+grade
│   (avg_price, median_price, min_price, max_price, sale_count, volume_bucket)
├── population_reports  — PSA/CGC/BGS/SGC pop snapshots
├── sentiment_raw       — individual Reddit/Twitter observations
├── sentiment_scores    — rolled-up 24h/7d/30d sentiment scores
├── feature_store       — pre-computed ML features per card
├── model_predictions   — model outputs with confidence bands
│   (fair_value, p10, p25, p50, p75, p90, buy_threshold, sell_threshold,
│    confidence [HIGH/MEDIUM/LOW], volume_bucket)
├── price_alerts        — triggered alerts (price_spike, price_crash, viral_social,
│                         anomaly_detected, new_high, new_low)
└── ingestion_log       — pipeline run tracking (source, status, records_processed, error_message)

Archival: old observations moved to R2 (Parquet/CSV)
D1 limit: 10GB per database — plan for archival at ~6 months
```

1.4 Data Pipeline Architecture

The entire pipeline runs on **Cloudflare Cron Triggers**, **Cloudflare Queues**, and **Durable Object agents**. There is no Airflow, no separate orchestrator.

```
                    ┌──────────┐
                    │  Sources │
                    └──────────┘
                         │
      ┌──────────────────┼──────────────────┐
      │                  │                  │
eBay/SoldComps    Reddit scrape      PriceCharting
(every 15 min)    (every 5 min,     (daily 2am)
                   rotates subreddits)
      │                  │                  │
      ▼                  ▼                  ▼
┌──────────────────────────────────────────┐
│     Cloudflare Cron Triggers + Queues    │
│                                          │
│  */15 * * * *  → SoldComps ingestion     │
│  */5  * * * *  → Reddit sentiment        │
│  0   * * * *   → Sentiment rollup (hourly) │
│  0 2 * * *     → PriceCharting           │
│  0 3 * * *     → PSA population reports  │
│  0 4 * * *     → Anomaly detection       │
│  0 5 * * *     → Aggregates + features   │
│  0 6 * * *     → Batch predictions       │
│                                          │
│  Queue: gamecards-ingestion (async batch) │
│  Queue: gamecards-sentiment (async AI)   │
└──────────────────────────────────────────┘
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                    ▼
┌──────────────────────────────────────────┐
│       Cloudflare D1 (Price History)      │
│  + KV (cache hot prices, rate limiting)  │
│  + R2 (raw data archive, model artifacts) │
│  + Workers AI (sentiment NLP — Gemma 4 26B) │
└──────────────────────────────────────────┘
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          │         │         │
          ▼         ▼         ▼
   Pricing API  Dashboard  Agents (DOs)
   (Hono/Workers) (Vite/React) (4 agents)
```

Pipeline ordering is critical — anomaly detection (4 AM) must run before features/predictions (5-6 AM) so flagged outliers are excluded from downstream computation.

2. Feature Engineering

2.1 Feature Categories & Importance

Based on research across collectibles pricing systems, academic papers, and production implementations (PriceCharting, Alt, Card Ladder, StockX):

RANK	FEATURE GROUP	PREDICTIVE POWER	IMPLEMENTATION COMPLEXITY
1	**Grade + Grading Company**	~25-30% of variance	Low — structured data from PSA/CGC
2	**Card Identity** (set, number, player/character)	~20-25%	Medium — learned embeddings
3	**Recent Sold Prices** (lagged features)	~15-20%	Low — rolling stats from price history
4	**Population Report** (supply)	~8-12%	Medium — daily pop scraping
5	**Sold Velocity** (demand signal)	~5-8%	Low — count sales per window
6	**Macro Market Index**	~3-5%	Medium — aggregate market health
7	**Player/Meta Performance**	~3-5%	High — domain-specific APIs
8	**Seasonality**	~2-3%	Low — calendar features
9	**Social Sentiment**	~1-3%	High — NLP pipeline required
10	**Rarity Metrics**	~2-4%	Medium — print run data

2.2 Critical Feature Details

Grade & Population Features:

```
{
    'grade_numeric': 9.5,                   # 1-10 scale
    'grading_company': 'PSA',               # PSA, BGS, CGC, SGC
    'is_gem_mint': True,                    # grade ≥ 9.5
    'is_perfect_10': False,
    'pop_at_grade': 47,                     # how many exist at this grade
    'pop_higher': 3,                        # how many graded higher
    'pop_ratio': 0.047,                     # % of total pop at this grade
    'is_pop_1': False,                      # unique at this grade
    'pop_growth_rate_90d': 0.12,            # 12% more graded in last 90 days
    # KEY INSIGHT: PSA 10 with pop 5 is worth dramatically more than pop 500
    # The relationship is highly nonlinear — tree models capture this well
}
```

Demand Signals: `sales_count_7d/30d/90d`, `velocity_trend` (7d/30d ratio), `price_momentum` (7d MA / 30d MA).

> **DEFERRED features** (require live listing snapshots): `active_listings_count`, `days_of_inventory`, `sell_through_rate`, `avg_bidders_last_5_auctions`. Need eBay Browse API + listing-snapshot pipeline.

Sentiment Features (from Reddit pipeline): `social_mention_count_7d`, `social_mention_trend` (7d/30d ratio), `social_sentiment_score` (-1 to 1), `reddit_post_count_7d`, `influencer_mention_7d`, `search_interest_trend`.

Seasonality Features: `is_holiday_season`, `is_tax_refund_season`, `is_summer_lull`, `is_nfl_season`, `days_since_set_release`, cyclical month encoding (`month_sin`, `month_cos`).

GameStop-Exclusive Features (the moat) — DEFERRED: `gamestop_trade_in_volume_7d`, `gamestop_store_price`, `gamestop_inventory_count`, `gamestop_days_in_inventory`, `gamestop_regional_demand`. No competitor has this data — requires internal DB integration.

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3. ML Model Architecture

3.1 The Core Problem: Heterogeneous Liquidity

Not all cards are created equal. A base set Charizard PSA 10 sells multiple times per week. A 1987 Topps Barry Bonds PSA 8 sells once every 2 months. The model architecture must adapt to volume.

3.2 Volume-Aware Ensemble

```
┌──────────────────────────────────────────────────────────┐
│                    ENSEMBLE PRICING ENGINE                 │
├──────────────────────────────────────────────────────────┤
│                                                            │
│  HIGH VOLUME (50+ sales/quarter)           ~15% of cards    │
│  ├── LightGBM quantile regression (primary)                 │
│  ├── Temporal Fusion Transformer (temporal dynamics)         │
│  └── LSTM for momentum capture                              │
│  → Meta-learner weighted average                            │
│  → Tight confidence intervals (±10-15%)                     │
│                                                            │
│  MEDIUM VOLUME (10-50 sales/quarter)       ~25% of cards    │
│  ├── LightGBM with stronger regularization                  │
│  ├── Gaussian Process regression (natural uncertainty)       │
│  └── KNN comparable sales (cold-start backup)               │
│  → GP uncertainty widens intervals (±15-25%)                │
│                                                            │
│  LOW VOLUME (< 10 sales/quarter)           ~60% of cards    │
│  ├── Hierarchical Bayesian model (borrows from similar)     │
│  ├── GP with informative priors                             │
│  └── Comparable sales (KNN in feature space)                │
│  → Wide uncertainty bands (±25-40%), flag for human review  │
│                                                            │
│  MARKET INDEX (feeds into all above as a feature)           │
│  └── Prophet decomposition on aggregate market data         │
│                                                            │
│  META-LEARNER                                               │
│  └── Stacking regressor: learns which base model to trust   │
│      based on volume, price range, volatility               │
└──────────────────────────────────────────────────────────┘
```

3.3 Primary Model: LightGBM Quantile Regression

LightGBM is the right primary model because: - Handles mixed feature types natively (categorical grades, continuous prices) - Captures nonlinear grade-price relationships (PSA 9 vs 10 can be 5-10x) - Fast inference (<1ms per card) - Handles missing features gracefully (many features will be NaN for low-volume cards) - Interpretable via SHAP values

```python
import lightgbm as lgb

# Train on log(price) — normalizes fat-tailed distribution
params = {
    'objective': 'huber',           # Overridden to 'quantile' in the loop below
    'metric': 'mae',
    'learning_rate': 0.03,
    'num_leaves': 63,
    'min_data_in_leaf': 20,
    'feature_fraction': 0.8,
    'bagging_fraction': 0.8,
    'bagging_freq': 5,
    'lambda_l1': 0.1,
    'lambda_l2': 1.0,
    'verbose': -1,
}

# Quantile models for uncertainty — includes p20/p80 for buy/sell thresholds
quantiles = [0.10, 0.20, 0.25, 0.50, 0.75, 0.80, 0.90]
models = {}
for q in quantiles:
    params_q = {**params, 'objective': 'quantile', 'alpha': q}
    models[q] = lgb.train(params_q, train_set, num_boost_round=2000,
                          valid_sets=[val_set],
                          callbacks=[lgb.early_stopping(50)])

# Inference: full price distribution
fair_value = np.exp(models[0.50].predict(X))      # Median = fair market value
buy_threshold = np.exp(models[0.20].predict(X))    # Buy if below this
sell_threshold = np.exp(models[0.80].predict(X))   # Sell if above this
confidence_width = (np.exp(models[0.90].predict(X)) - np.exp(models[0.10].predict(X))) / fair_value
```

Why Huber loss, not MSE: A single anomalous $3,000 sale of a card that normally trades at $50 will destroy a model trained with MSE. Huber loss is robust to these fat-tailed outliers.

Why log(price): Collectibles prices are log-normally distributed. A $50 card varying +/-$10 is the same relative uncertainty as a $5,000 card varying +/-$1,000.

3.4 Conformal Prediction for Calibrated Uncertainty

Quantile regression gives intervals, but they may not be calibrated (the 90% interval might only cover 78% of actuals). Conformal prediction wraps any model to give **distribution-free coverage guarantees** by computing nonconformity scores on a calibration set and using the empirical quantile as the interval width.

3.5 Buy/Sell Decision Framework

> **Critical:** Thresholds are based on **net realizable value (NRV)** — what GameStop actually nets after selling — not raw market quantiles.

Economics constants (implemented in `PricingRecommendationAgent`): marketplace fee 13%, shipping $5, return rate 3%, required margin 20%. NRV = `fair_value * (1 - fee) * (1 - return_rate) - shipping`. Max buy price = `NRV * (1 - required_margin)`.

4. Sentiment Analysis Pipeline

4.1 Architecture

```
Reddit via Browser Rendering (old.reddit.com/.json)
        │
        ▼
┌──────────────────────────────────────┐
│  NLP Pipeline (Workers AI)           │
│                                      │
│  1. Card Mention Extraction (NER)    │
│     "That Charizard PSA 10 is fire"  │
│     → card: Charizard, grade: PSA 10 │
│                                      │
│  2. Sentiment Classification         │
│     Workers AI (Gemma 4 26B +        │
│     DistilBERT) at the edge          │
│     → bullish (0.85)                 │
│                                      │
│  3. Hype Detection                   │
│     Volume spike + positive sentiment │
│     → HYPE_ALERT: Charizard PSA 10   │
│                                      │
│  4. Aggregation (hourly cron)        │
│     Rolling 24h / 7d / 30d scores    │
│     Weighted by engagement (upvotes) │
└──────────────────────────────────────┘
                    │
                    ▼
         D1 sentiment_scores table
         → Feeds into ML feature pipeline
```

Note: Reddit deprecated its Data API in favor of Devvit, so we scrape via Cloudflare's headless Chrome (Browser Rendering) instead of using the API directly. We fetch `old.reddit.com/<subreddit>.json` through Browser Rendering, which returns structured JSON without requiring API credentials.

Note: Twitter/X integration is **DEFERRED**. v1 uses Reddit only.

4.2 Target Subreddits

SUBREDDIT	MEMBERS	FOCUS	EST. DAILY VOLUME
r/PokemonTCG	~280K	Pokemon cards	~500 posts/day
r/baseballcards	~150K	Baseball	~300 posts/day
r/basketballcards	~60K	Basketball	~150 posts/day
r/footballcards	~50K	Football	~120 posts/day
r/PKMNTCGDeals	~100K	Pokemon deals/pricing	~200 posts/day
r/sportscards	~30K	General sports	~80 posts/day
r/TradingCardCommunity	~15K	General TCG	~50 posts/day

4.3 Viral Event Detection

Social media sentiment's biggest value is **catching viral moments** before they hit prices. The **PriceMonitorAgent** implements this: if social mentions in a 6-hour window exceed 3x the 7-day rolling average, it fires a `viral_social` alert and invalidates the KV price cache for affected cards.

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5. Anomaly Detection

5.1 Seller-Side Anomaly Detection

Data limitation: SoldComps, CardHedger, and PriceCharting return sold-listing summaries, not bid-level event data. True shill-bidding detection is not feasible without a bid-history feed.

```
What we CAN detect from sold listings:
├── Seller price inflation — sellers consistently above market (via seller_id if available)
├── Statistical price outliers — IQR-based detection on price vs. historical average
├── Suspicious pricing patterns — high-value graded cards at <$1 (data quality)
└── Price spike/crash detection — 7d vs 30d moving average divergence > 30%

What we CANNOT detect without bid-event data:
├── Same bidder repeatedly losing to same seller
├── Bid increment manipulation
├── Concentrated bidding in final seconds
└── Buyer-seller relationship graphs

Model: Statistical outlier detection + seller concentration (if seller_id available)
Action: Flag price_observations.is_anomaly = true, exclude from training/features
```

Anomaly detection runs daily at **4 AM UTC** (before features at 5 AM, predictions at 6 AM) so flagged outliers are excluded from downstream computation.

5.2 Data Quality Issues

- **"Best Offer Accepted" bias:** eBay shows listing price, not accepted price. Biases upward 15-25%. Fix: flag and discount by 20%, or exclude.
- **Lot sales:** Filter via keyword detection (lot, bundle, collection, set of, x2, x3).
- **Currency normalization:** International sales in GBP/EUR/AUD must be converted to USD at the sale date exchange rate.

6. Backtesting & Evaluation

6.1 Walk-Forward Validation

Never use random train/test splits for time series data. Always train on past, test on future. Full walk-forward requires 6+ months of raw transaction data. At launch, use PriceCharting backfill with reduced feature set.

6.2 Metrics

METRIC	TARGET
MdAPE (Median Absolute % Error)	<15% high-vol, <25% mid-vol, <40% low-vol
Directional Accuracy	>60%
Coverage (90% CI)	87-93%
Interval Width %	<30% for high-vol
Simulated Trading P&L	Positive over backtest

Critical: Report metrics **stratified by volume bucket.**

7. System Architecture & Infrastructure

7.1 Technology Stack

LAYER	TECHNOLOGY	STATUS
Runtime	Cloudflare Workers (Hono framework)	**Shipped**
Data Storage	Cloudflare D1 (SQLite at edge)	**Shipped**
Cache	Cloudflare KV (hot price lookups, rate limiting)	**Shipped**
Object Storage	Cloudflare R2 (data archive, model artifacts)	**Shipped**
Queues	Cloudflare Queues (gamecards-ingestion, gamecards-sentiment)	**Shipped**
Orchestration	Cloudflare Cron Triggers (8 schedules)	**Shipped**
Agents	Cloudflare Durable Objects (Agents SDK)	**Shipped** — 4 agents
NLP	Workers AI (Gemma 4 26B + DistilBERT)	**Shipped**
Browser Rendering	Cloudflare headless Chrome	**Shipped** — Reddit scraping without API
ML Training	LightGBM + scikit-learn (offline, Python)	**Shipped**
ML Export	ONNX via onnxmltools/skl2onnx → R2	**Shipped**
ML Serving	Batch predictions (R2 JSON → D1 model_predictions)	**Shipped**
Experiment Tracking	MLflow (local/self-hosted)	**Shipped**
Frontend	Vite + React + Tailwind v4 + React Router	**Shipped**
Auth	API key middleware on `/v1/*` + agent routes	**Shipped**
CI	GitHub Actions (typecheck, lint, Python syntax)	**Shipped**
Retraining	GitHub Actions weekly cron (Sunday 2 AM UTC)	**Shipped**
Monitoring	Cloudflare Analytics + `ingestion_log` table	**Shipped**
A/B Testing	Model comparison framework	**DEFERRED**
GameStop Integration	Internal DB for trade-in/inventory data	**DEFERRED**

7.1.1 Security & Authentication

Implemented: - **SignIn gate:** Dashboard requires an access code entry on a SignIn page. The code is server-validated against the `DEMO_ACCESS_CODE` secret. - **Session tokens:** `POST /v1/auth/login` accepts `{ code }` and returns an HMAC-signed session token with 24-hour expiry, stored in KV. - **Auth middleware:** All `/v1/*` routes accept either a static API key (`X-API-Key` header) OR a session token (`Authorization: Bearer <token>`). Bypassed in development environment. - **Session management:** `POST /v1/auth/verify` checks token validity and returns `{ valid, expiresAt }` . `POST /v1/auth/logout` invalidates the session token in KV. - **Agent auth:** All `/agents/*` routes also gated by the same auth middleware in non-development environments. - **CORS:** Locked to allowed origins, configurable via `ALLOWED_ORIGINS` environment variable. - **Rate limiting:** KV-based throttling at 120 requests/min per token on `/v1/*` . - **Secrets management:** All API keys stored via `wrangler`

`secret put` — `SOLDCOMPS_API_KEY`, `PRICECHARTING_API_KEY`, `CARDHEDGER_API_KEY`, `POKEMON_PRICE_TRACKER_KEY`, `API_KEY`, `DEMO_ACCESS_CODE`, `SESSION_SECRET`. - **D1 access:** Restricted to Workers bindings only — no public database endpoint.

Required before production: - **GameStop SSO:** Dashboard should require GameStop SSO for production. - **Data privacy:** `seller_id` should be hashed before storage.

7.1.2 Custom Domains & Deployment

- **API:** `api.gmestart.com` — Cloudflare Worker (Hono)
- **Dashboard:** `app.gmestart.com` — Cloudflare Pages (Vite + React)
- **SignIn gate:** Dashboard requires access code `GMESTART2026` before granting a session token. The code is validated server-side via `POST /v1/auth/login`.

7.2 API Endpoints

All endpoints are under `/v1/` and require API key or session token authentication (except the root health check and auth endpoints).

Auth:

```
POST /v1/auth/login              — { code } → { token, expiresAt }
POST /v1/auth/verify             — → { valid, expiresAt }
POST /v1/auth/logout             — → { status }
```

Root Health Check:

```
GET  /
     → { service, version, status: "healthy"|"degraded", environment,
         auth, predictions: "fresh"|"stale" }
     Checks model_predictions freshness (stale if >36 hours old).
```

Cards:

```
GET  /v1/cards                   — List/search card catalog
GET  /v1/cards/:id               — Get single card details
```

Prices:

```
GET  /v1/price/:card_id?grade=&grading_company=
     → { card_id, card_name, grade, grading_company, price, lower, upper,
         buy_threshold, sell_threshold, confidence, last_sale, sales_30d,
         trend, updated_at, has_prediction }
     Returns model prediction if available, statistical fallback otherwise.
```

History:

```
GET  /v1/history/:card_id?grade=&days=
     → [{ date, price, source, sale_type }, ...]
```

Sentiment:

```
GET  /v1/sentiment/:card_id
     → { score, mention_count, period, top_posts, computed_at }
```

Evaluate (Buy/Sell Decision):

```
POST /v1/evaluate
     Body: { card_id, offered_price, grade?, grading_company? }
     → { decision: "STRONG_BUY"|"REVIEW_BUY"|"FAIR_VALUE"|"SELL_SIGNAL",
         fair_value, margin, confidence, reasoning }
```

Alerts:

```
GET  /v1/alerts                  — Active alerts
POST /v1/alerts/:id/resolve      — Resolve an alert
```

Market:

```
GET  /v1/market/index
     → { pokemon_index, sports_index, trend_30d, volatility, updated_at }
```

System (operational):

```
GET  /v1/system/health           — Pipeline health (prediction freshness, model version,
                                   ingestion status, catalog count)
GET  /v1/system/model            — Current model metadata from R2
POST /v1/system/rollback         — Full rollback: copies versioned predictions from R2,
                                   re-imports into D1, invalidates KV cache, updates meta
     Body: { version_key: "models/versions/..." }
POST /v1/system/bootstrap        — Bootstrap card catalog from PriceCharting
POST /v1/system/seed             — Seed database with sample data for development/demo
POST /v1/system/mock-internal    — Generate mock internal GameStop data for testing
```

Agents (REST proxy for Durable Object @callable methods):

```
GET  /v1/agents/monitor/status
POST /v1/agents/monitor/check

GET  /v1/agents/intelligence/latest
POST /v1/agents/intelligence/generate

GET  /v1/agents/competitors/status
GET  /v1/agents/competitors/gaps
POST /v1/agents/competitors/scan

GET  /v1/agents/recommendations/pending?action=BUY|SELL|REPRICE
GET  /v1/agents/recommendations/status
POST /v1/agents/recommendations/generate
POST /v1/agents/recommendations/:id/approve   Body: { approvedBy? }
POST /v1/agents/recommendations/:id/reject    Body: { rejectedBy? }
```

Agents are also accessible via WebSocket using the Agents SDK `useAgent()` hook from the dashboard.

7.3 Autonomous Agents (Durable Objects)

Four Durable Object agents run autonomously alongside the cron pipeline. Each uses the Cloudflare Agents SDK with hibernation, automatic retry, and real-time state sync via WebSocket.

7.3.1 PriceMonitorAgent

Purpose: Detects price anomalies and viral social events in real time, triggers cache invalidation for affected cards.

PROPERTY	VALUE
Schedule	`scheduleEvery(900)` — every 15 minutes
Hibernate	Yes
Retry	3 attempts, 1-30s backoff
Max alerts	100 (validated on state change)

State: `lastCheckAt`, `activeAlerts[]`, `checksRun`, `anomaliesDetected`

Callable methods: - `runMonitoringCheck()` — Queries D1 for 30% 1d-vs-30d price divergence + viral social mentions (>3× 7d average in 6h). Writes `price_alerts` rows. Invalidates KV cache for affected cards. - `getStatus()` — Returns last check time, alert counts, recent alerts. - `clearAlerts()` — Clears active alerts. - `listSchedules()` / `cancelTask(id)` — Schedule management.

7.3.2 MarketIntelligenceAgent

Purpose: Generates daily AI-powered market briefings using Workers AI (Gemma 4 26B).

PROPERTY	VALUE
Schedule	`schedule("0 7 * * *")` — daily at 7 AM
Hibernate	Yes
Retry	2 attempts, 5-60s backoff
Reports stored	Last 30

State: `reports[]`, `lastGeneratedAt`, `totalReports`

Callable methods: - `generateDailyReport()` — Queries D1 for top gainers/decliners (7d), active alerts, sentiment summary, volume stats. Sends to Gemma 4 26B for prose summary. Returns `MarketReport` with highlights, top movers, and market sentiment (bullish/bearish/neutral). - `getLatestReport()` — Most recent report. - `getReportHistory(count)` — Last N reports (default 7). - `getStatus()` — Generation stats.

7.3.3 CompetitorTrackerAgent

Purpose: Scans for price gaps between GameStop's fair value and competitor prices (PriceCharting, CardHedger, SoldComps).

PROPERTY	VALUE
Schedule	`scheduleEvery(21600)` — every 6 hours
Hibernate	Yes
Retry	3 attempts, 2-30s backoff
Max gaps tracked	50

State: `lastScanAt`, `priceGaps[]`, `scansCompleted`, `opportunitiesFound`

Callable methods: - `scanCompetitorPrices()` — Compares top 100 cards' `model_predictions.fair_value` against recent `price_observations` from external sources. Flags gaps >15%. Matches on grade+grading_company to avoid false gaps. - `getOverpriced(limit)` / `getUnderpriced(limit)` — Filtered gap lists. - `getAllGaps()` — All gaps sorted by magnitude. - `getStatus()` — Scan stats with overpriced/underpriced counts.

7.3.4 PricingRecommendationAgent

Purpose: Generates actionable BUY/SELL/REPRICE recommendations with human-in-the-loop approval workflow.

PROPERTY	VALUE
Schedule	`schedule("0 8 * * *")` — daily at 8 AM (recommendations)
	`scheduleEvery(21600)` — every 6h (expire stale)
Hibernate	Yes
Retry	2 attempts, 5-60s backoff
Max pending	200 (validated on state change)

State: `pending[]`, `history[]`, `lastGeneratedAt`, `stats` (totalGenerated/Approved/Rejected/Expired)

Economics constants: `MARKETPLACE_FEE = 13%`, `SHIPPING_COST = $5`, `RETURN_RATE = 3%`, `REQUIRED_MARGIN = 20%`

Callable methods: - `generateRecommendations()` — For all HIGH/MEDIUM confidence predictions with fair_value > $10: computes NRV, generates BUY (market < max buy price), SELL (market > sell threshold), or REPRICE (near breakeven). Top 50 by margin, sorted. - `approveRecommendation(id, approvedBy)` — Moves to history as approved. - `rejectRecommendation(id, rejectedBy)` — Moves to history as rejected. - `getPending(action?)` — Filter pending by action type. - `getHistory(limit)` — Approval/rejection history. - `getStatus()` — Pending counts by action, stats. - `expireStaleRecommendations()` — Auto-expires pending recs older than 48 hours.

8. Implementation Roadmap

Phase 1: Foundation — DONE

DELIVERABLE	STATUS
D1 schema + migrations	Done
SoldComps API integration (every 15 min)	Done
PriceCharting API integration (daily)	Done
PSA population scraping (daily)	Done
Feature engineering pipeline	Done
LightGBM quantile regression training pipeline	Done
ONNX export + R2 upload	Done
Batch scoring CLI	Done

Phase 2: Intelligence — DONE

DELIVERABLE	STATUS
Reddit sentiment pipeline (NER + classification via Workers AI)	Done
Sentiment rollup (24h/7d/30d, hourly cron)	Done
Anomaly detection (daily, before features)	Done
Conformal prediction intervals	Done
Volume-aware routing (high/medium/low buckets)	Done
Buy/sell decision API (`POST /v1/evaluate`)	Done
Hono/Workers API serving layer with auth	Done

Phase 3: Production — DONE

DELIVERABLE	STATUS
React dashboard (market overview, card detail, search, evaluate, alerts)	Done
Pipeline health monitoring (`/v1/system/health` , `ingestion_log`)	Done
Model rollback system (`POST /v1/system/rollback`)	Done
Automated weekly retraining (GitHub Actions, Sunday 2 AM)	Done
CI pipeline (typecheck, lint, Python syntax check)	Done
Price alerts system (spike, crash, viral, anomaly, new high/low)	Done
KV price cache with automatic invalidation	Done

Phase 4: Agentic Layer — DONE

DELIVERABLE	STATUS
PriceMonitorAgent (anomaly + viral detection, every 15 min)	Done
MarketIntelligenceAgent (daily AI market briefing)	Done
CompetitorTrackerAgent (price gap scanning, every 6h)	Done
PricingRecommendationAgent (BUY/SELL/REPRICE with approval workflow)	Done
Agent REST proxy + WebSocket access	Done
Agent dashboard page in frontend	Done

Remaining Work

ITEM	PRIORITY	NOTES
Twitter/X sentiment integration	P2	~$500-1K/mo, filtered stream
GameStop internal data integration (trade-ins, inventory, foot traffic)	P1	Requires internal DB access
A/B testing framework (model comparison)	P2	Compare model versions on live traffic
Model drift detection + automated alerts	P2	Track prediction accuracy over time
CORS lockdown to GameStop domains	P0	Currently `cors("*")`
GameStop SSO for dashboard	P1	Production auth
eBay Marketplace Insights API access	P1	Negotiate via retail partnership
TCGPlayer API integration	P2	Requires application approval
Hierarchical Bayesian / GP models for low-volume	P3	LightGBM with fallback is v1
Temporal Fusion Transformer for high-volume	P3	Research item
D1 archival strategy implementation	P2	Needed at ~6 months

9. Competitive Landscape

	PRICECHARTING	CARD LADDER (PSA)	ALT	
Data sources	eBay + own marketplace	14 platforms	Multiple	SoldComps + PriceCharting + PSA pop + Reddit
ML pricing	Algorithmic smoothing (no ML)	Unknown	Likely gradient boosting	LightGBM quantile regression (batch) + statistical fallback
Sentiment	None	None	Unknown	Reddit NLP (Workers AI)
Update frequency	Daily	Near real-time	Near real-time	Hybrid: 15-min ingestion + daily model + 15-min agent monitoring
Uncertainty	None	None	None	Conformal prediction intervals (p10-p90)
Physical retail data	No	No	No	**DEFERRED** — requires internal integration
Anomaly detection	Manual review	Unknown	Unknown	Automated (IQR outliers, data quality, viral detection)
Autonomous agents	No	No	No	4 agents (monitor, intelligence, competitors, recommendations)
Buy/sell decisions	No	No	Unknown	NRV-based with margin targets + human approval workflow

Honest assessment: v1 does not yet have GameStop's "unfair advantage" (internal store data). The moat exists but is not yet plumbed. What ships is a best-in-class ML pricing engine with uncertainty quantification and autonomous monitoring — competitive with or ahead of Alt/Card Ladder on technical sophistication, behind on data breadth until internal data is connected.

10. Cost Estimate (Cloudflare Stack)

Monthly Operating Costs

ITEM	COST/MONTH	NOTES
Cloudflare Workers Paid plan	$5	Includes 10M requests/mo
Cloudflare D1	~$5-15	Based on rows read/written
Cloudflare KV	~$5-10	Price cache + rate limiting
Cloudflare R2	~$5-10	Model artifacts, data archive (no egress fees)
Cloudflare Queues	~$1-5	Two queues, moderate volume
Cloudflare Workers AI	~$10-50	Gemma 4 26B for reports + DistilBERT for sentiment
Durable Objects (4 agents)	~$5-15	Billed per request + storage
SoldComps API (Scale)	$59	Primary market data
CardHedger API	$49	Multi-platform data
PriceCharting (Legendary)	~$25	Historical/aggregated prices
PokemonPriceTracker (Business)	$99	Graded Pokemon data
Cloudflare Browser Rendering	~$5-10	Reddit scraping via headless Chrome
GitHub Actions (retraining)	~$0-10	Free tier usually sufficient
MLflow (local)	$0	Self-hosted
Total	**~$280-$360/month**	

Deferred costs (not in v1): - Twitter/X API: ~$500-1K/mo - ML training compute (if moved to cloud): ~$20-50/mo - GameStop internal data pipeline: $0 (internal infra)

vs. Palantir/vendor alternative: $100K-500K+/year for a fraction of the functionality.

11. Key Risks & Mitigations

RISK	SEVERITY	MITIGATION
eBay API access denied	High	SoldComps + CardHedger as fallbacks; negotiate via GameStop's retail partnership
Model accuracy insufficient for low-volume cards	Medium	Volume-aware routing, wide uncertainty bands, human review for <10 sales/quarter
Social sentiment is noisy/manipulable	Low	Weight sentiment low (1-3% of signal), use as regime detector not price predictor
Price history cold start	Medium	Backfill from PriceCharting CSV, SoldComps 365-day lookback
Fat-tailed prices break the model	Medium	Huber loss, log-transform, quantile regression, IQR outlier detection
eBay "Best Offer" bias (+15-25%)	Medium	Detect and discount by 20%, or exclude from training
Reddit scraping reliability	Low	Browser Rendering may break if old.reddit.com changes format; fallback to direct .json endpoint or Devvit app
D1 10GB database limit	Medium	Archive old price_observations to R2 at ~6 months; keep only recent windows in D1
Worker CPU time limits (30s)	Medium	Batch D1 writes at 90 statements; chunk large operations; use Queues for async
KV eventual consistency	Low	5-minute TTL on price cache; evaluate endpoint reads D1 directly
Worker 128MB memory limit	Low	Batch predictions loaded from R2 JSON; paginate for >100K cards
Durable Object single-point-of-failure	Low	Agents use hibernation + retry with backoff; state persists across restarts
R2 model artifact corruption	Medium	Versioned predictions in `models/versions/`; full rollback system implemented
Cron trigger overlap	Low	Pipeline is ordered with 1-hour gaps; each stage is idempotent
CORS currently open (`*`)	High	Must lock to GameStop domains before production

Appendix A: References & Data Sources

APIs & Data: - eBay Marketplace Insights API - SoldComps — eBay Sold Listings API - CardHedger Enterprise API - PriceCharting API - PSA Public API - GemRate — Aggregated Pop Reports - PokemonPriceTracker API - Reddit API Pricing 2026

Research & Methodology: - PriceCharting Methodology - CAIA: Trading Cards and the Price of Perfection - eBay Bans AI Agents — Feb 2026

Appendix B: ML Training CLI

The `packages/ml-training` Python package provides these CLI commands:

COMMAND	PURPOSE
`gmestart-export-features`	Export features + training data from D1 via HTTP API
`gmestart-train`	Train LightGBM quantile models
`gmestart-export`	Export trained models to ONNX format
`gmestart-backtest`	Walk-forward backtesting
`gmestart-score`	Batch score all cards, upload predictions to R2

Automated retraining: GitHub Actions workflow runs weekly (Sunday 2 AM UTC): export features from D1 → train → ONNX export → batch score → upload predictions to R2 → Worker picks up within 10 minutes.

Appendix C: Frontend Pages

ROUTE	COMPONENT	PURPOSE
`/`	MarketOverview	Action queue with market stats + recent alerts
`/search`	SearchBar	Card search with category filtering
`/card/:cardId`	CardDetail	Price history, predictions, sentiment for a card
`/evaluate`	EvaluateCard	Buy/sell/hold recommendation at a given price
`/alerts`	AlertsList	Alert triage with resolve controls
`/agents`	AgentDashboard	Real-time agent status via WebSocket

Categories: All Cards, Pokemon, Baseball, Basketball, Football, MTG, Yu-Gi-Oh.

Built with production experience from Exmplr (multi-agent AI platform over 562K+ clinical trials, RAG over 19.8M vectors) and ServeSys.AI (healthcare edge AI, 27K LOC Rust, 384 tests). The same architectural patterns — deterministic recipe routing, evaluation frameworks, quality gates, incremental pipelines — apply directly to collectibles pricing.